Sample Papers

CS402 Theory of Automata

Mid Term Examination – Spring 2006

Time Allowed: 90 Minutes

Please read the following instructions carefully before attempting any question:

1. **This examination is closed book, closed notes, closed neighbors.**
2. **Answer all questions.**
   a. **There is no choice.**
   b. **You will have to answer correctly all questions in this examination to get the maximum possible marks.**
3. **Do not ask any questions about the contents of this examination from anyone.**
   a. **If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.**
   b. **If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.**
4. **You are allowed to use any Software for Diagrams and Symbols like MS Word, MathType and Visio etc.**

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**WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.

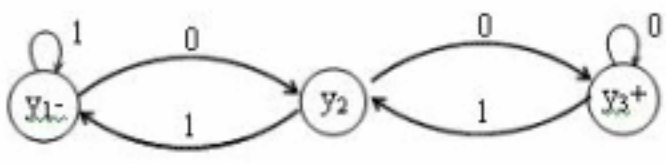
Question No. 1 **Marks : 1**

If s=abcd is a string defined over $\sum$ = {a,bc,d} then reverse of s is dcba.

★ True
★ False

Question No. 2 **Marks : 10**

Find the regular expression associated to the following FA. Show all steps.
[Hint: u



Question No. 3 **Marks : 1**

$\sum$ = {aa, b}, length(aaaabaabb) = 5.

★ True
★ False

Question No. 4 **Marks : 1**

Every NFA can be converted into FA.

★ True
★ False

Question No. 5 **Marks : 1**

There can be more than one start states in TG.

★ True
★ False

Question No. 6 **Marks : 1**

A regular language can not be infinite.

★ True
★ False

Question No. 7 **Marks : 10**

a) Write the recursive definition of the following language. [6]

L = Defining the language $\{a^{2n}b^{4n}\}$, n=1,2,3,... , of strings defined over Σ={a, b}

b) Write a regular expression of the language having strings that either start or end with "00" and have no more zeroes. Where the alphabet is {0, 1}. [4]

Question No. 8 **Marks : 1**

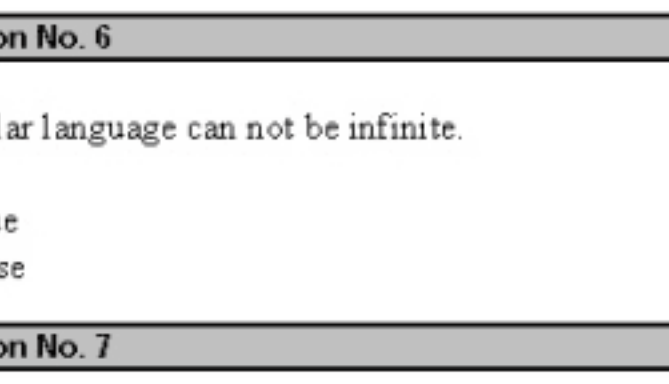
Kleene star of {1} generates {1, 11, 111, 1111, 11111 ……}.

★ True
★ False

Question No. 9 **Marks : 10**

a) Define NFA-null. [4]

b) Draw DFA for the following NFA. [6]



Question No. 10 **Marks : 1**

If a regular language is empty then we denote it like L = Ø (fi).

★ True
★ False

Question No. 11 **Marks : 1**

Recursive method for defining language is only for regular languages.

★ True
★ False

Question No. 12 **Marks : 1**

$aa^* = a^+$?

★ True
★ False

Question No. 13 **Marks : 1**

The language equal means number of a's and b's are equal with null string.

★ True
★ False